

Original Spin $\times$ SU(7) action

Nine independent fields; fixed background and original time

Local nine-field development

Seven homogeneous initial data
All nine Euler residuals vanish
Actual derivative e^{tJ_0}

J

Coupled propagation

289 original field coordinates
103-dimensional faithful quotient
79 prepared + 24 complementary

Spatial matter evolution

$L^2(\mathbb{R}^3, \mathbb{C}^{12})$; control
All real times and amplitudes
Unitary evolution $W_\varepsilon(t, a)$

Original matter readout

Complete products on $\mathcal{C}^{252}$
Independent dual $\chi = s\psi^\dagger S$
Density $a^3\rho = 4\sqrt{2}$

Actual matter exchange

158 vertices; on-shell sources
Original kinetic residues
Reconstruct all 289 equations

Prepared current

$K^\dagger AK$; every finite CAR word
 $B = -\sqrt{2} QT$
Two-sided expectation

The same response at $\varepsilon = 0$

Derivative of $W =$ Duhamel; current derivative = Kubo = source readout